

AST425 Project Report: Distinguishing Polarizations with the Next Generation of Gravitational Wave Detectors

STUART WILLIAMSON¹ AND REED ESSICK²

¹*Department of Astronomy and Astrophysics, University of Toronto, Toronto, Ontario M5S 1A7*

²*Canadian Institute for Theoretical Astrophysics, 60 St. George St, Toronto, Ontario M5S 3H8*

ABSTRACT

The next generation of gravitational wave detectors will have arms an order of magnitude larger than LIGO. The long-wavelength regime typically assumed of current detectors will no longer apply, resulting in more complicated responses to signals. We utilize this feature to investigate and quantify the distinguishability of gravitational wave polarizations. For networks of 3 to 4 detectors, signals with SNRs similar to those in the current catalogue can have their polarization determined with a likelihood ratio of 10. Such a network would enable testing of general relativity, which predicts only two polarization modes, versus beyond-GR theories of gravity which allow an additional 4 modes.

1. INTRODUCTION

1.1. Gravitational Wave Polarizations

Probing the polarization content of gravitational waves (GW) provides a direct test of General Relativity (GR). GR predicts that only two polarizations of GWs exist, while more general theories of gravity predict up to 6 (Eardley et al. 1973). These polarizations are labelled and categorized as tensor (+, ×), vector (x, y), and scalar (b, ℓ). GR allows only the tensor modes of polarization, + and ×. Evidence of non-tensor GW polarization modes or a lack of their observation provide evidence for or against beyond-GR theories of gravity.

1.2. Current and Future Detectors

Current generation detectors typically operate in the low-frequency regime. For a GW of frequency f and a detector of arm length L , the low-frequency limit applies when $f \ll \frac{c}{2L} := f_{\text{FSR}}$. When this is true, the GW amplitude is $\approx$ constant during the time it takes light to travel down the interferometer arm and back ($\frac{2L}{c}$). Interferometer-based GW detectors measure the phase difference between light reflected up and down each detector arm. In the low-frequency limit then, the light experiences the length of the detector arm as about constant. The phase difference between arms can thus be directly taken as an instantaneous measurement of the differential arm length. The GW amplitude follows simply from that. LIGO's 4km arms have $f_{\text{FSR}} \sim 37.5\text{kHz}$. For typical astrophysical signals of $f \lesssim 2\text{kHz}$ this is well within the low-frequency regime (Essick et al. 2017).

In this work we consider detectors of the same L-shaped design as LIGO but with 40km arms. Cosmic

Explorer is such a proposed detector (Evans et al. 2021). This size of detector has $f_{\text{FSR}} \sim 3.75\text{kHz}$. Thus the low-frequency approximation is no longer valid and a detailed treatment of the detector response is required for high frequency signals. Essick et al. (2017) does this analysis for the tensor modes + and ×, providing the mathematical basis for **gw-detectors**³ which simulates the high-frequency response of detectors from any polarization. Essick & Isi (2023) consider the scalar modes b and ℓ at high-frequency. They find promising results for distinguishing them despite the opposite being true in the low-frequency regime. In this work we consider all six polarizations in both regimes and analyze their distinguishability.

1.3. Matched Filtering

We use the process of matched filters given in Essick (2023, Sec. II) to compare the effect of different polarizations on detector response. This is a method to detect the presence of a signal template $h(f)$ in some data $d(f)$. It analytically maximizes the amplitude (SNR) of h that best matches the data as well as the phase ϕ across templates of the form $h = h_0 e^{i\phi}$.

For a single detector, the likelihood $p(d|h)$ of observing some data d , supposing the signal was of shape h , is given by the formula

$$\ln p(d|h) = -2 \int_0^\infty \frac{|d(f) - h(f)|^2}{S(f)} df \quad (1)$$

³ <https://git.ligo.org/reed.essick/gw-detectors>

Where $S(f)$ is the one-sided power spectral density (PSD) of the detector noise (Essick 2023, eq. 8).

This leads to the matched filtering process of calculating

$$\rho(h) = \left| \frac{4}{\mathcal{N}(h)} \int_0^\infty \frac{d^* h}{S} df \right| \quad (2)$$

where $\mathcal{N}(h) = \sqrt{4 \int_0^\infty |h|^2 / S df}$ is a normalization factor. This calculation analytically maximizes over both the amplitude and the phase of h (Essick 2023). By comparing many different templates against d , the template with the greatest $\rho(h)$ has the highest likelihood of being the true signal.

This can be extended to a network of detectors by summing $\rho(h)$ in quadrature, i.e.

$$\rho^{\text{net}}(d; h) = \left(\sum_j \rho(d^j; h^j)^2 \right)^{1/2} \quad (3)$$

where d^j, h^j are the results of projecting signals d, h onto each detector (Essick 2023, eq. 16). Note that previously, d and h were already projected onto the singular detector.

By generating a set of templates $H = \{h_i\}$ that span the potential signal's parameter space, finding the maximum $\rho_H^{\text{net}}(d) = \max_i \{\rho^{\text{net}}(d; h_i)\}$ gives the signal parameters most likely to have created the data d .

In section 2, we give the methodology of our work. In section 3, we present the findings made using our procedure and interpret their implications. In section 4 we summarize our results and discuss avenues for further work on the project.

2. PROCEDURE

To test the ability of next-gen detectors to distinguish polarizations, we simulate the response of detector networks made of one to four 40km detectors. We create a ‘true’ signal and attempt to recover the signal parameters using matched filtering of ‘test’ signals. To do this we use the **gw-detectors** python package developed by Prof. Essick. Along with modifications for efficiency we make use of code snippets from Jewel Cao’s modifications to the package.⁴

2.1. Network and Signal Configuration

While **gw-detectors** allows for a range of detector configurations, we stick with two perpendicular arms. The PSD used (S in sec 1.3) is the Cosmic Explorer design PSD that comes with the package. The location

and arm orientation otherwise fully defines a detector. We test a variety of networks, composed of hypothetical detectors dispersed across the globe, including replacing the Hanford LIGO site in Washington, US. and the Virgo site in Italy with 40m detectors.

For the signal morphology, we use a sine-Gaussian with a quality factor of 6. We consider two central frequencies $f_0 = 0.1 f_{\text{FSR}}$ and $f_0 = 0.85 f_{\text{FSR}}$ to represent the low- and high-frequency regimes respectively.

The remaining quantities that need to be specified to project the signal onto the detector network are: polarization, location on the sky, time of arrival, and polarization angle. For polarization, do not consider linear combinations of polarizations for simplicity and clarity of results. Instead we consider only purely polarized signals in this work. Location on the sky is simply the source of the signal in spherical or celestial coordinates, (θ, ϕ) or (α, δ) . Time of arrival t corresponds to the phase of each bin in frequency space when the signal reaches the center of the Earth. The polarization angle ψ relates the wave frame x - and y -axes to the detector frame’s, with it being the simple rotation required when the signal is directly overhead the detector.

To create our ‘true’ signal we then simply choose values for each of these parameters. Since we are investigating polarization, we will consider all six cases separately using the same parameters otherwise for each. We choose a few representative signal locations to consider, spread across the sky so we can spot and avoid any atypical behaviour (e.g. the signal is nearly overhead a detector). For time of arrival and polarization angle, we set each to 0 in the code as these are cyclic and relative parameters that have no inherent meaning.

By projecting this signal onto the detector network we get our data d_j^γ for each detector j and each polarization $\gamma \in \{+, \times, x, y, b, \ell\} := P$.

2.2. Maximizing Parameters

Next, we compare the constructed ‘true’ signals of each polarization to a set of templates, finding the maximum. We again consider each polarization case separately, now giving a total of 36 pairs of true and ‘test’ polarizations $(\gamma, \beta) \in P$. For each β we create the template bank H_β and calculate $\rho_{\gamma, \beta} = \rho_{H_\beta}^{\text{net}}(d_\gamma)$ by directly maximizing over the parameters $\theta \in (0, \pi)$, $\phi \in (0, 2\pi)$, $\psi \in (0, \pi)$.

The maximization over t is done using an inverse fast Fourier transform (iFFT) on the integrand of equation (2). We want to calculate $\rho(h)$ using the templates $h(t; f) = e^{2\pi i t f} h(f)$ for each $h(f)$ given by the other parameters θ, ϕ, ψ . Computing the integral of equation (2) for each $h(t; f)$ is equivalent to computing the iFFT

⁴ https://github.com/charrybdis/gw-detectors_modified

of d^*h/S . This is significantly more efficient as it avoids projecting the signal onto the network many times.

2.3. Comparing Polarizations

To compare the distinguishability of polarization pairs we use likelihood ratios. To be able to tell apart some polarization γ from another polarization β , we want $p(d_\gamma|h_\gamma)$ to be significantly larger than $p(d_\gamma|h_\beta)$ and for $p(d_\beta|h_\beta)$ to be significantly larger than $p(d_\beta|h_\gamma)$. If this is the case then a γ polarized signal is badly described by a β polarized signal and vice-versa, meaning the polarizations are observationally distinguishable. To quantify the distinguishability, we investigate the likelihood ratio $\Lambda = \ln p(d_\gamma|h_\gamma) / \ln p(d_\gamma|h_\beta)$.

We can convert a desired likelihood ratio Λ , say $\Lambda = 10$, into a threshold SNR beyond which polarizations are distinguishable. From eq. 1 we find that, after maximizing over templates to get $\rho_{\gamma,\beta}$ along with the maximizing template h_β , $\ln p(d_\gamma|h_\beta) = \frac{1}{2}\rho_{\gamma,\beta}^2$. Thus, $\Lambda = \exp(\frac{1}{2}\rho_{\gamma,\gamma} - \frac{1}{2}\rho_{\gamma,\beta})$ and we can rearrange this into

$$\rho_{\gamma,\beta}^{\text{thr}} = \sqrt{\frac{2 \log \Lambda}{1 - (R_{\gamma,\beta})^2}} \quad (4)$$

where $R_{\gamma,\beta} = \rho_{\gamma,\beta} / \rho_{\gamma,\gamma}$. The interpretation of $\rho_{\gamma,\beta}^{\text{thr}}$ is the minimum SNR of a γ polarized signal for which we select the hypothesis that it is in polarization γ rather than β with likelihood ratio Λ .

By comparing $\rho_{\gamma,\beta}^{\text{thr}}$ with SNR bounds of current GW detections we can probe the ability of next-gen detectors to distinguish polarizations.

2.4. Multi-Messenger Astronomy

In the event of an associated electromagnetic observation with the GW detection, such as GW170817 (Abbott et al. 2017), the source location is precisely known. The advantage gained from this scenario is that it removes the degeneracy between some polarizations and the orientation relative to a detector. For example, with a single detector an x polarized GW from directly overhead is indistinguishable from an ℓ polarized wave along the x axis. To consider this case we use the same procedure as described above, simply considering only templates with the true (θ, ϕ) . Results of this are presented in section 3.2.

3. RESULTS

We tested networks of 1, 2, 3, and 4 detectors in 6 configurations that span 7 locations around the world. For each configuration we tested signals of 6 different source locations. Our results were broadly similar in each case with only minor details changing. A representative configuration was chosen that has features shared by the

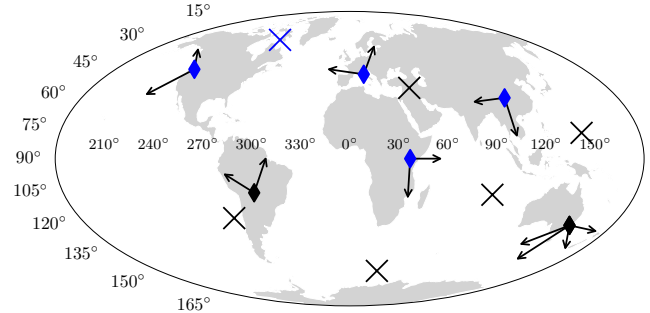


Figure 1. Detectors and signal locations tested. Detectors are marked by diamonds with arm directions shown by arrows, signals are marked with crosses. Blue markers show the configuration for which figures 2 and 3 correspond to, with detectors added to the network west-to-east. There are two sets of arrows on the Australian detector as we tested networks with both rotations of this detector's arms.

majority of network and signal combinations. All detectors and signal locations considered are visualized in figure 1, with the representative configuration highlighted. This network consists of detectors at the Hanford LIGO site in the United States, the Virgo site in Italy, and hypothetical sites in Kenya and in China. The signal location is in the North of Canada.

3.1. Threshold SNR

Figure 2 shows the result of calculating $\rho_{\gamma,\beta}^{\text{thr}}$ from equation (4) for the detector network and signal highlighted in figure 1. The plot is coloured in three bins: 0-10, 10-43, and 43 or above. The first bin corresponds to SNRs which are near or below the detection threshold. This means that any signal we reasonably believe is not noise will meet the distinguishability threshold $\rho_{\gamma,\beta}^{\text{thr}}$. The third bin contains signals with exceptionally high SNR. With the exception of GW250114 (Abac et al. 2025) which had SNR of near 80, the loudest signal in the GWTC-4.0 catalogue had an SNR of 42.1 (LIGO Scientific, VIRGO, KAGRA Collaborations et al. 2025). Thus the third bin corresponds to polarizations which are indistinguishable by reasonable signal observations. The second bin, containing SNRs 10-43, thus corresponds to polarizations which may be distinguishable for some signals but not others, depending on the signal strength. White squares are where either the ratio $R_{\gamma,\beta}$ is unity to numerical precision or is greater than 1. Both of these cases cause equation 4 to be undefined and they both mean that the two polarizations are indistinguishable (or that $\gamma = \beta$ and there is no meaning to $\rho_{\gamma,\beta}^{\text{thr}}$ anyway).

A feature of figure 2 common among detector configurations is the lack of distinguishability of any polarizations in the low-frequency regime ($f_0 = 0.1 f_{\text{FSR}}$).

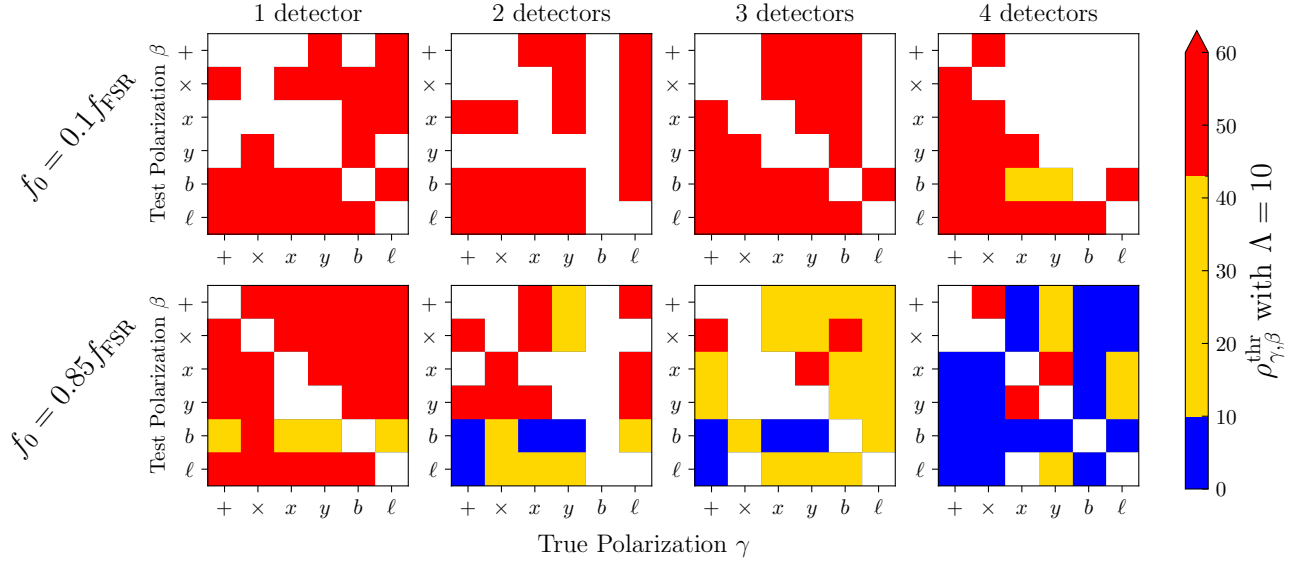


Figure 2. Visualization of the results of equation (4) for the network-signal configuration highlighted in figure 1. The detectors are added West-to-East as shown in figure 1. Colouring is grouped into bins corresponding to: always distinguishable (0-10, blue), sometimes distinguishable (10-43, yellow), and never distinguishable (43 & above, red) [see sec 3.2]. White squares represent where $\rho_{\gamma,\beta}^{\text{thr}}$ is undefined and also represent indistinguishability when not on the diagonal.

This confirms that current generation detectors are incapable of distinguishing polarizations. Another expected feature is the increasing distinguishability of polarizations with the number of detectors in the network in the high-frequency regime. We also expect that $+$ is always indistinguishable from $\times$ and that x is from y . This is because they only differ by a z -axis rotation of $\pi/2$ and π respectively in the wave-frame. Our maximization over ψ causes this indistinguishability and corresponds to the fact that we choose the coordinate system that defines the polarizations. All of these features are seen in across the detector configurations as well as figure 2, confirming the basis of our knowledge.

We now begin an analysis of the specifics of figure 2. A similar interpretation of the other network configurations can be done, reaching broadly the same conclusions. We can see that complete distinguishability of the polarizations (in the high-frequency regime) begins only upon adding the third detector to the network. With only two detectors the scalar modes are potentially distinguishable but not quite, while with three it only depends on signal strength to distinguish most polarizations. Using the four-detector network, all polarizations are distinguishable with only some requiring loud signals, other than the expected pairs $+/ \times$ and x/y .

Taking a specific example to illustrate, consider the b polarization in the high-frequency 4-detector case. The fact that the entire b -column is blue shows that a detectable b polarized GW can always be confirmed as an b polarized GW over any other polarization mode.

Thus if such a signal existed we would be able to support beyond-GR physics. Conversely, the entire b -row is also blue. This means that for a signal of any other polarization, we could rule out the possibility of it being b polarized, potentially supporting GR as a theory. This kind of reasoning applied to the rest of figure 2 gives the clear picture that next-gen detectors are clearly able to distinguish polarizations in a network of 4 detectors, potentially with 3 detectors for loud enough signals, and in certain cases even with just 2 detectors.

3.2. Multi-Messenger Threshold SNR

As discussed in section 2.4, multi-messenger astronomy can provide a precise signal location for us to match to. The result of calculating $\rho_{\gamma,\beta}^{\text{thr}}$ from equation (4) in this case using the same detector network and signal as highlighted in figure 1 and used in figure 2 is presented in figure 3.

We can immediately see that, apart from the diagonal, there are no longer any undefined values of $\rho_{\gamma,\beta}^{\text{thr}}$ as in figure 2. This is likely because knowing the signal location removes numerical precision issues that arise from scanning across the sky as a grid.

Further comparing to figure 2, the performance of in the high-frequency is broadly much better as predicted in section 2.4. With 4 detectors all polarizations are completely distinguishable where possible, and nearly so with 3 detectors. Even with only 1 or 2 detectors it is possible to make conclusions about polarizations such as ruling out the ℓ polarization in signals differently polarized.

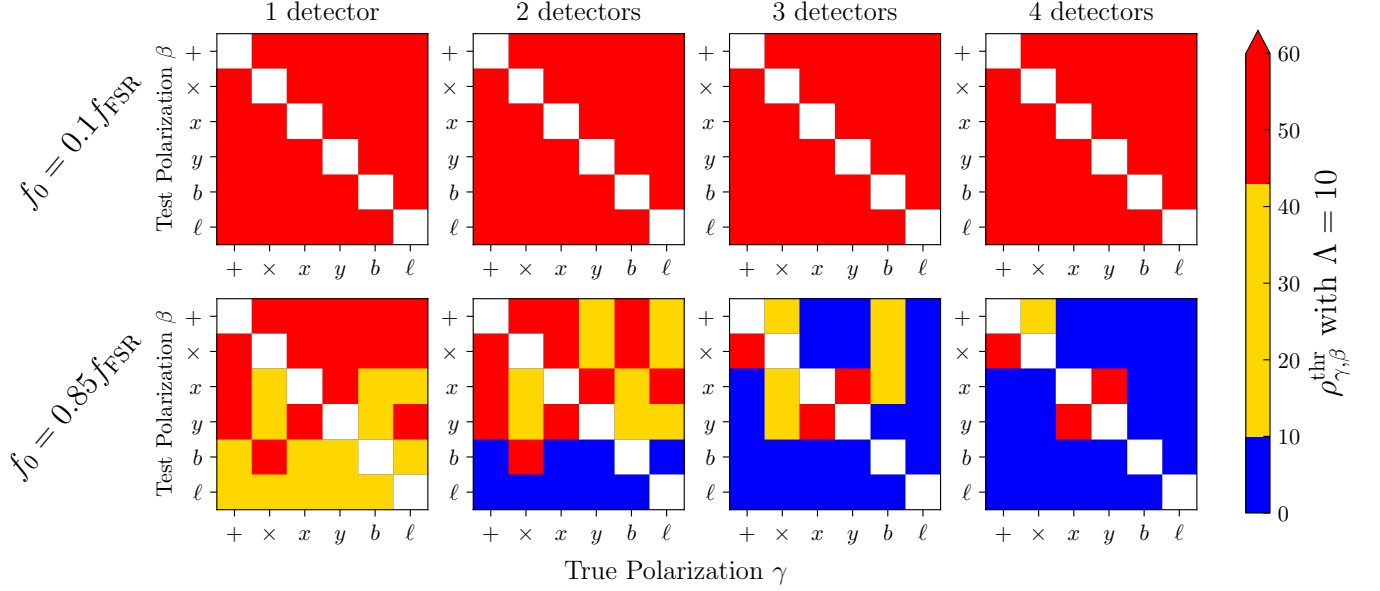


Figure 3. Visualization of the results of equation (4) for the network-signal configuration highlighted in figure 1 when the location of the signal is known. See the caption of figure 2 for an explanation of the figure and section 3.2 for its interpretation.

4. CONCLUSION

We have shown that next-generation detectors show significant promise for distinguishing polarizations with even relatively few detectors. Especially when paired with electromagnetic observations, GW detectors with 40km arms can differentiate signals of different polarizations for typical astrophysically motivated frequencies with confidence. The effectiveness of a network increases with number of detectors. Distinguishability begins to be feasible at three detectors in general. With an associated electromagnetic observation, distinguishing polarizations is possible with only one or two detectors in some cases. The ability to distinguish signal polarizations can be used to support GR and also to suggest beyond-GR physics through the detection or inobservability of vector and scalar polarization modes. These

results are consistent across multiple locations of detectors across the globe as well as when varying the source location of the signal.

Future work on this project could take multiple directions. Performing a more exhaustive and systematic search over source signal location would help make a concrete analysis of the effectiveness of an individual detector network. Similarly, a systematic method of selecting and comparing detector placement on the globe could inform ideal GW detector locations. These paths are likely prohibitively computationally expensive using the methods presented here, so finding optimizations in our procedure could be highly useful. One such optimization is restricting the (θ, ϕ) search to analytically computed triangulation rings.

REFERENCES

- Abac, A. G., Abouelfettouh, I., Acernese, F., et al. 2025, Phys. Rev. Lett., 135, 111403, doi: [10.1103/kw5g-d732](https://doi.org/10.1103/kw5g-d732)
- Abbott, B. P., Abbott, R., Abbott, T. D., et al. 2017, Phys. Rev. Lett., 119, 161101, doi: [10.1103/PhysRevLett.119.161101](https://doi.org/10.1103/PhysRevLett.119.161101)
- Eardley, D. M., Lee, D. L., & Lightman, A. P. 1973, Phys. Rev. D, 8, 3308, doi: [10.1103/PhysRevD.8.3308](https://doi.org/10.1103/PhysRevD.8.3308)
- Essick, R. 2023, Phys. Rev. D, 108, 043011, doi: [10.1103/PhysRevD.108.043011](https://doi.org/10.1103/PhysRevD.108.043011)
- Essick, R., & Isi, M. 2023, Distinguishing scalar gravitational-wave polarizations with next-generation detectors, Cosmic Explorer Science Letter. https://dcc.cosmicexplorer.org/public/0163/P2300008/001/Scalar_polarizations_in_3G.pdf
- Essick, R., Vitale, S., & Evans, M. 2017, Physical Review D, 96, doi: [10.1103/physrevd.96.084004](https://doi.org/10.1103/physrevd.96.084004)
- Evans, M., Adhikari, R. X., Afle, C., et al. 2021, A Horizon Study for Cosmic Explorer: Science, Observatories, and Community. <https://arxiv.org/abs/2109.09882>
- LIGO Scientific, VIRGO, KAGRA Collaborations, et al. 2025. <https://arxiv.org/abs/2508.18082>